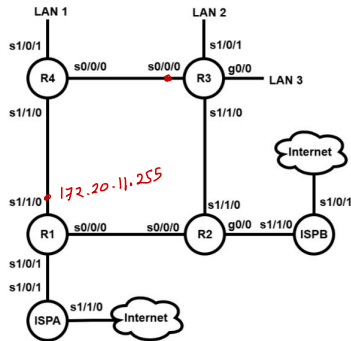


Q3

Device	Interface	IP	Network
R1	s0/0/0	.101	172.20.10.100/30
	s1/1/0	.225	172.20.11.224/30
	s1/0/1	.1	172.20.9.0/30
R2	s1/1/0	.193	172.20.12.192/30
	s0/0/0	.102	172.20.10.100/30
	g0/0	.1	178.51.100.0/30
R3	s1/0/1	.1	172.20.96.0/24
	g0/0	.1	172.20.80.0/26
	s1/1/0	.194	172.20.12.192/30
R4	s0/0/0	.97	172.20.10.96/30
	s1/0/1	.1	172.20.64.0/25
	s1/1/0	.226	172.20.11.224/30
ISPA	s1/0/1	.2	172.20.9.0/30
ISPB	s1/1/0	.2	178.51.100.0/30
	s1/0/1	.1	215.1.1.0/24



- [CO3] I. On R4, configure a **static default route (recursive)** toward the upstream via R1.
- [CO3] II. Configure a **back-up route (directly attached)** of the above static route (AD 5).
- [CO3] III. On R3, add a **recursive** static route to reach the WAN link between ISPB and R2.
- [CO2] IV. What inefficiency will the static route in (III) create?
- [CO2] V. Two default static routes are configured on R2: S* 0.0.0.0/0 [10/0] via 178.51.100.2 and S* 0.0.0.0/0 [20/0] via 172.20.10.101. Which entry should be in the routing table of R2, and why?

I. ip route 0.0.0.0 0.0.0.0 172.20.11.225

II. ip route 0.0.0.0 0.0.0.0 s0/0/0 5

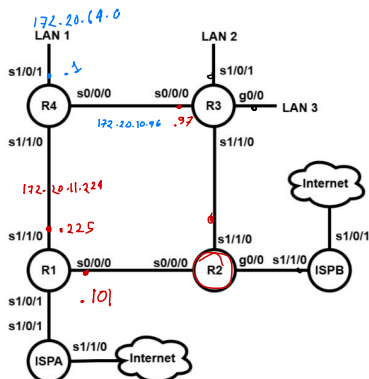
III. ip route 178.51.100.0 255.255.255.252 178.20.12.193

IV. Recursive lookup consumes extra CPU cycles.

V. As AD 10 < AD 20, which is more trustworthiness
so R2 config 178.51.10.2 as primary path.

Q3

Device	Interface	IP	Network
R1	s0/0/0	.101	172.20.10.100/30
	s1/1/0	.225	172.20.11.224/30
	s1/0/1	.1	172.20.9.0/30
R2	s1/1/0	.193	172.20.12.192/30
	s0/0/0	.102	172.20.10.100/30
	g0/0	.1	178.51.100.0/30
R3	s1/0/1	.1	172.20.96.0/24
	g0/0	.1	172.20.80.0/26
	s1/1/0	.194	172.20.12.192/30
R4	s0/0/0	.97	172.20.10.96/30
	s1/0/1	.1	172.20.64.0/25
	s0/0/0	.98	172.20.10.96/30
ISPA	s1/1/0	.226	172.20.11.224/30
	s1/0/1	.2	172.20.9.0/30
ISPB	s1/1/0	.1	210.1.1.0/24
	s1/0/1	.2	178.51.100.0/30
ISPB	s1/0/1	.1	215.1.1.0/24
	s1/0/1	.1	215.1.1.0/24

3
+
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+

- [CO3] I. On R2, add a **recursive** static route to LAN1 via R1.
- [CO3] II. Configure a **back-up route (directly attached)** of the above static route (AD 6).
- [CO3] III. On R4, add a **recursive default** static route to go through the ISPB link.
- [CO2] IV. If the above route wasn't configured on R4, what problem would have arisen?
- [CO2] V. Two static routes are configured on R3: S 172.20.64.0/25 [30/0] via 172.20.10.98 and S 172.20.64.0/25 [20/0] via 172.20.12.193. Which entry should be in the routing table of R3, and why?

1. ip route 172.20.64.0 255.255.255.128 172.20.10.101

11. ip route 172.20.64.0 255.255.255.128 s1/1/0 6

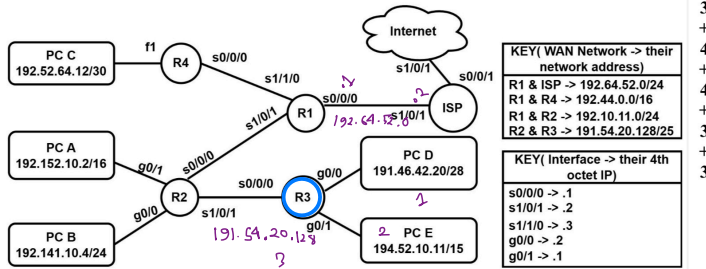
111. ip route 0.0.0.0 0.0.0.0 172.20.10.97

1V. Router 4 drop unknown packets.

1V. Use lower AD as more trustworthiness.

S 172.20.64.0/25 [20/0] via 172.20.12.193

Q2



- [CO2] I. On R3, identify the networks that will be added to the table without any routing configuration.
- [CO3] II. Configure a directly attached static route on R2 to reach R4 LAN with AD 50
- [CO3] III. Configure a recursive route so that a S* entry is added in R1's routing table. Identify the significance of S*.
- [CO2] IV. Determine the AD of a back up route for III.

(Please Turn Over) 1

A

A

- [CO2] V. R4 is discarding the packet whose destination IP is 172.42.10.4. Deduce the reason behind this.

I. R3 will be connected with PC D, PC E and R2 to R3
191.46.42.20, 194.52.10.11, 191.54.20.128

II. Directly attached:

	IP	subset mask	exit int.	AD
ip route	192.52.64.12	255.255.255.252	s0/0/0	50

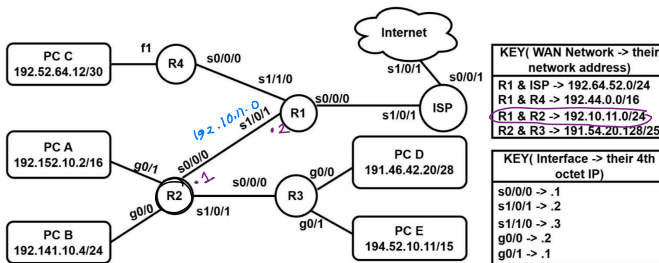
III. Recursive Route: Default S* [static default]

	IP	subset	Next hop IP [192.64.52.0] [2.2]
ip route	0.0.0.0	0.0.0.0	192.64.52.2

IV. For static route default AD is 1. So, for backup route AD will be greater than 1.

V. Destination IP does not belong to any network in topology.

Q2



- [CO2] I. From the routing table, mention how you can identify the directly connected networks
- [CO3] II. Configure a recursive default static route on R2.
- [CO3] III. Configure a directly attached floating default static route in R2 for the above if R3 is connected to another ISP using interface s0/0/1.
- [CO2] IV. Determine the significance of [40/0] in the routing table.
- [CO2] V. Why is a directly attached static route better than a recursive route?

I. By c (connected) and L (Local)

II. ip route 0.0.0.0 0.0.0.0 192.10.11.2

III. ip route 0.0.0.0 0.0.0.0 s1/0/1 ⁵ AD > 1

IV. AD = 40 and cost = 0

V. Lower cpu overhead and Interface dependency, faster lookup.